CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A solid sphere has radius 21 cm. The volume of the sphere is: 1

(a) 19404 cm³ **(b)** 38808 cm³ **(c)** 5544 cm³ **(d)** 29106 cm³

2. For the data with classes 10–20, 20–30, 30–40, 40–50 with frequencies 5, 8, 20, 17 respectively, The mean of the data is: 1

(a) 35 **(b)** 34.8 **(c)** 35.8 **(d)** 435

3. The probability of an event E is $\frac{1}{4}$. The probability of the event 'not E' is: 1

(a) $\frac{5}{4}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{3}{4}$

4. A solid hemisphere has radius 7 cm. The total surface area of the hemisphere is: 1

(a) 462 cm² **(b)** 308 cm² **(c)** 154 cm² **(d)** 616 cm²

5. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 3, 9, 4, 18, 16 respectively, The mode of the data is: 1

(a) 60 **(b)** 50 **(c)** 52 **(d)** 58.75

6. The probability of a sure event is: 1

(a) 0 **(b)** 2 **(c)** $\frac{1}{2}$ **(d)** 1

7. A right circular cylinder has base radius 14 cm and height 10 cm. The curved surface area of the cylinder is: 1

(a) 880 cm² **(b)** 2112 cm² **(c)** 1760 cm² **(d)** 6160 cm²

8. For the data with classes 0–10, 10–20, 20–30, 30–40 with frequencies 3, 9, 5, 8 respectively, The median of the data is: 1

(a) 20 **(b)** 30 **(c)** 21 **(d)** 22.2

9. A die is thrown once. The probability of getting an odd number is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{3}{7}$

10. A right circular cylinder has base radius 14 cm and height 5 cm. The volume of the cylinder is: 1

(a) 1672 cm³ **(b)** 6160 cm³ **(c)** 3080 cm³ **(d)** 440 cm³

11. The upper limit of a class 20–30 is: 1

(a) 30 **(b)** 20 **(c)** 10 **(d)** 25

12. A die is thrown once. The probability of getting a perfect square is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{3}$

13. A right circular cone has base radius 14 cm and height 48 cm. The slant height of the cone is: 1

(a) 50 cm **(b)** 62 cm **(c)** 31 cm **(d)** 47 cm

14. The class interval is 20–30. The class mark is: 1

(a) 20 **(b)** 25 **(c)** 10 **(d)** 30

15. A coin is tossed two times. The probability of getting at least one head is: 1

(a) 1 **(b)** $\frac{1}{4}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{2}$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A coin is tossed two times. Find the probability of getting two heads. 2

17. A cuboid measures 8 cm by 5 cm by 3 cm. Find the total surface area of the cuboid. 2

18. The marks obtained by students in a test are given below. 2

Marks	40–50	50–60	60–70	70–80
Number of students	4	10	4	7

Find the median.

19. A bag contains 5 red, 3 blue balls. One ball is drawn at random. Find the probability that it is blue. 2

20. A right circular cylinder has base radius 14 cm and height 6 cm. Find the total surface area of the cylinder. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	10–20	20–30	30–40	40–50	50–60
Number of patients	2	10	2	2	9

Find the mode.

22. A class has 40 students, of whom 24 are girls. A student is selected at random to be the monitor. Find the probability that a boy is selected. 3

23. A right circular cone has base radius 14 cm and height 48 cm. Find the volume of the cone. 3

24. The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	0–10	10–20	20–30	30–40	40–50
Number of workers	8	15	8	8	11

Find the mean.

OR

In a lottery there are 8 prizes and 32 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

25. A right circular cone has base radius 14 cm and height 48 cm. Find the total surface area of the cone. 3

OR

The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	50–60	60–70	70–80	80–90	90–100
Number of patients	5	17	5	18	5

Find the median.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The distances (in km) travelled to office by employees of a company are given below. **5**

Distance (km)	10–20	20–30	30–40	40–50	50–60
Number of employees	10	2	6	16	6

Find the median.

OR

A box contains 12 cards numbered 1 to 12. One card is drawn at random. Find the probability that the number is: **5**

(i) a number less than 10,

(ii) a prime number,

(iii) a number greater than 12 (in this order).

27. A solid metal cuboid 24 cm by 12 cm by 12 cm is melted and recast into cubes of edge 6 cm. Find the number of cubes obtained. **5**

OR

The daily wages (in ₹) of workers in a factory are given below. **5**

Daily wage (₹)	10–20	20–30	30–40	40–50	50–60
Number of workers	2	16	10	4	8

Find the mean.

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND